

Analysis of Melanoma Cell Type in Uveal Melanoma Following Treatment Failure

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- **PURPOSE:** To determine whether an increase (or progression) in the degree of malignancy according to cytologic or histopathologic criteria corresponded with failure of intraocular melanoma control.
- **DESIGN:** Retrospective case control study from a single institution.
- **METHODS:** Uveal melanoma patients had either fine-needle aspiration biopsy followed by irradiation and later enucleation or an eyewall resection followed by enucleation. The observation procedures were serial histology or cytopathology and histology, reanalyzed in a masked manner. The main outcome measures were change in the predominant tumor cell type, local tumor control, and metastases.
- **RESULTS:** There was not a significant correlation between a change in melanoma cell type and failure to control intraocular melanoma. Similarly, the latency between treatment and enucleation was actually less (but not statistically so) in tumors that showed no increase in melanoma malignancy cell type compared with those that had a more malignant cell type on serial examination.
- **CONCLUSIONS:** There was no correlation between a change to more a malignant cell type and local control failure in patients treated either with radiation or eyewall resection for uveal melanoma. (Am J Ophthalmol 2004;138:543–546. © 2004 by Elsevier Inc. All rights reserved.)

CUTANEOUS MELANOMA PROGRESSION FROM A benign pigmented lesion to a metastatic tumor has been extensively characterized.^{1,2} In contrast, there is a scarcity of data on the cellular evolution of uveal melanomas either before or after various forms of therapy.³ We and others have demonstrated that posttreatment intraocular tumor recurrence is associated with higher tumor-related mortality.^{4–6} Is the progression of melanoma

cell type partially responsible for this observation? It is conceivable that either spindle cell or mixed cell melanomas, when they recur, could become more malignant with a greater percentage of epithelioid cells in the tumor. Bechrakis and associates analyzed 15 uveal melanomas that had local resections and later required enucleation.⁷ In 10 of the cases, recurrent tumors showed a more malignant cell type. Recent studies show that epithelioid compared with spindle cell melanomas exhibit a more malignant molecular behavior.^{8–11}

We have previously published on fine-needle aspiration biopsy (FNAB) in uveal melanomas.¹² No adverse effect was noted with this procedure, and in irradiated cases it established an additional independent prognostic parameter.¹³ In fine-needle biopsy cases followed by immediate enucleation, there was an excellent correlation (> 95% agreement) between cell type based on FNAB vs that independently assessed with standard histopathology.^{12,13}

This study analyzed two uveal melanoma patient groups to determine whether alterations in melanoma cell types occurred after unsuccessful local treatment. Group 1 comprised patients who had fine-needle aspiration cytopathology before charged particle radiation or I-125 brachytherapy and later required enucleation. Group 2 comprised patients who had eyewall resections and later underwent enucleation.

This was a retrospective case control study from a single institution. Uveal melanoma patients had either FNAB followed by irradiation and later enucleation or an eyewall resection followed by enucleation. The observation procedures were serial histology or cytopathology and histology, reanalyzed in a masked manner. The main outcome measures were change in the predominant tumor cell type, local tumor control, and metastases.

METHODS

AFTER INSTITUTIONAL REVIEW BOARD APPROVAL AND retrospectively reviewing our uveal melanoma database, two patient groups were selected. The first included patients with irradiated uveal melanoma who had a preradiation FNAB and later required an enucleation, and the

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